Vaal University of Technology • Diploma in IT • AIBUY3A

TRIAGE AI: CLINICAL WORKFLOW OPTIMIZATION

An AI Solution for Healthcare Industries: Predictive Triage, Bed Allocation & Clinician Burnout Mitigation

Group: Triage AI

Theme: AI Solution for Industries

PROBLEM STATEMENT

Public Healthcare Crisis

- Overcrowded casualty & emergency departments with prolonged patient wait times.
- Clinicians lose 35% of shift hours to manual electronic health record documentation.
- Severe shortages of critical Intensive Care Unit (ICU) beds during admission surges.
- Preventable 30-day hospital readmissions drain municipal healthcare budgets.

PROJECT OBJECTIVE

Predictive Healthcare Architecture

- Machine Learning: 30-day readmission risk classification prior to discharge.
- Time Series Forecasting: SARIMAX models for emergency admission surges.
- Deep Learning Telemetry: PyTorch networks for continuous ICU vitals anomaly detection.
- Clinical NLP & Voice Dictation: Clinical note transcription and entity parsing.
- Operational Softbot: Real-time ward capacity and triage protocol queries.

System Architecture: 5 Core AI Disciplines

01

Machine Learning

Random Forest Classifier

Predicts 30-day readmission bounce-back risk

Scikit-Learn

02

Time Series

SARIMAX (1,1,1)

14-week emergency surge & bed volume forecasting

Statsmodels

03

Deep Learning

Vitals Multi-Layer Perceptron

Continuous ICU telemetry deterioration flagging

PyTorch

04

Clinical NLP

Entity Extraction & TTS

Voice dictation parsing & triage audio alerts

spaCy / pyttsx3

05

Operational Softbot

Conversational Assistant

Real-time queries on bed capacity & Code Blue

Rule-Based Intent

EMPIRICAL MODEL EVALUATION

Performance Metrics

Reported prototype results • supplied project figures

48.0%

Readmission Recall Rate

0.64

Model ROC-AUC Score

6.8 MAE

Admissions Forecast Error

94.2%

ICU Anomaly Accuracy

MUNICIPAL & COMMUNITY IMPACT

Local Healthcare Value

- Optimizes municipal health spending by preventing costly emergency re-hospitalizations.
- Targets up to 20% shorter emergency room wait times through surge staffing.
- Mitigates clinician burnout, returning critical hours to direct bedside patient care.
- Supports critical ICU bed availability for local community emergencies.

Intended benefits; local clinical validation required.

Triage AI Repository:
github.com/BB-Dube/Triage-AI

Developed for VUT Business Analysis 3.2
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